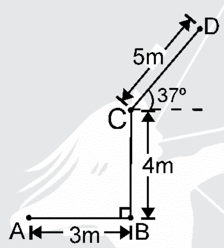
DPP No: 01 **Maximum Time** **50 Min**	competishun Power of real gurus **C³** ***(COMPETISHUN CRASH COURSE)***	***PHYSICS*** **TARGET** **JEE-MAINS**

SYLLABUS : KINEMATICS

1. A hall has the dimensions 10 m × 10 m × 10 m. A fly starting at one corner ends up at a farthest corner. The magnitude of its displacement is:

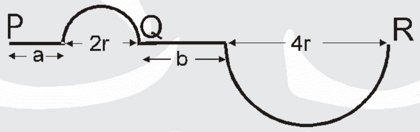
(A) $5\sqrt{3}$ m (B) $10\sqrt{3}$ m (C) $20\sqrt{3}$ m (D) $30\sqrt{3}$ m

2. A particle moves along a path ABCD as shown in the figure. Then the magnitude of net displacement of the particle from position A to D is :



(A) 10 m (B) $5\sqrt{2}$ m (C) 9 m (D) $7\sqrt{2}$ m

3. A car starts from P and follows the path as shown in figure. Finally car stops at R. Find the distance travelled and displacement of the car if a = 7 m, b = 8 m and r = $\frac{11}{\pi}$ m? [Take $\pi = \frac{22}{7}$]



(A) 15m, 36 m (B) 48m, 25 m (C) 35m, 28 m (D) 48m, 36 m

4. A car travels from A to B at a speed of 20 km h^{-1} and returns at a speed of 30 km h^{-1}. The average speed of the car for the whole journey is :

(A) 5 km h^{-1} (B) 24 km h^{-1} (C) 25 km h^{-1} (D) 50 km h^{-1}

5. A person travelling on a straight line without changing direction moves with a uniform speed v_1 for half distance and next half distance he covers with uniform speed v_2 . The average speed v is given by

(A) $v = \frac{2v_1 v_2}{v_1 + v_2}$ (B) $v = \sqrt{v_1 v_2}$ (C) $\frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$ (D) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$

6. A car covers a distance of 2 km in 2.5 minutes. If it covers half of the distance with speed 40 km/hr, the rest distance it shall cover with a speed of:

(A) 56 km/hr (B) 60 km/hr (C) 48 km/hr (D) 50 km/hr

7. A body covers first $\frac{1}{3}$ part of its journey with a velocity of 2 m/s, next $\frac{1}{3}$ part with a velocity of 3 m/s and rest of the journey with a velocity 6m/s. The average velocity of the body will be

(A) 3 m/s (B) $\frac{11}{3}$ m/s (C) $\frac{8}{3}$ m/s (D) $\frac{4}{3}$ m/s

8. A car runs at constant speed on a circular track of radius 100 m taking 62.8 s on each lap. What is the average speed and average velocity on each complete lap? (π = 3.14)

(A) velocity 10m/s, speed 10 m/s (B) velocity zero, speed 10 m/s

(C) velocity zero, speed zero (D) velocity 10 m/s, speed zero

9. A train travels 5 km in the north-east direction at the rate of 40 km h^{-1} and then 5 km at the rate of 40 km h^{-1} in the north–wast direction. What is its average speed during the whole journey and what is its average velocity.

(A) 15kmph, $\frac{40}{\sqrt{2}}$ kmph (B) 25kmph, $\frac{40}{\sqrt{2}}$ kmph (C) 40kmph, $\frac{40}{\sqrt{2}}$ kmph (D) 45kmph, $\frac{40}{\sqrt{2}}$ kmph

10. The displacement of a body is given by $2s = gt^2$ where g is a constant. The velocity of the body at any time t is:

(A) gt (B) gt/2 (C) $gt^2/2$ (D) $gt^3/6$

11. A truck travelling due north at 50 km/hr turns west and travels at the same speed. What is the change in magnitude of velocity.

(A) 1 (B) 2 (C) 5 (D) 0

12. Find the velocity as a function of time if $x = At + Bt^{-3}$, where A and B are constants, x is position and t is time.

(A) $A - 1\,Bt^{-4}$ (B) $A - 2\,Bt^{-4}$ (C) $A - 3\,Bt^{-4}$ (D) $A - 5\,Bt^{-4}$

13. An athelete takes 2s to reach his maximum speed of 18 km/h after starting from rest. What is the magnitude of his average accleration?

(A) $\frac{1}{2}$ m/s^2 (B) $\frac{5}{2}$ m/s^2 (C) $\frac{2}{5}$ m/s^2 (D) $\frac{3}{5}$ m/s^2

14. A particle performs rectilinear motion in such a way that its initial velocity has opposite direction with its uniform acceleration. Let x_A and x_B be the magnitude of displacements in the first 10 seconds and the next 10 seconds, then:

(A) $x_A < x_B$

(B) $x_A = x_B$

(C) $x_A > x_B$

(D) the information is insufficient to decide the relation of x_A with x_B.

15. A particle moves rectilinearly with a constant acceleration 1 m/s^2. Its speed after 10 seconds is 5 m/s. The distance covered by the particle in this duration is (Initial & final velocities are in opposite directio)

(A) 20 m (B) 25 m (C) 30 m (D) 50 m

16. A ball is dropped from the top of a building. The ball takes 0.5 s to fall past the 3 m height of a window some distance from the top of the building. If the speed of the ball at the top and at the bottom of the window are v_T and v_B respectively, then (g = 9.8 m/sec^2)

(A) $v_T + v_B = 12$ ms^{-1} (B) $v_T - v_B = 4.9$ m s^{-1} (C) $v_B v_T = 1$ ms^{-1} (D) $\frac{v_B}{v_T} = 1$ ms^{-1}

17. A stone is released from an elevator going up with an acceleration a and speed u. The acceleration and speed of the stone just after the release is

(A) a upward, zero (B) (g-a) upward, u

(C) (g-a) downward, zero (D) g downward, u

18. A particle is thrown upwards from ground. It experiences a constant air resistance force which can produce a retardation of 2 m/s^2 . The ratio of time of ascent to the time of descent is : [g = 10 m/s^2]

(A) 1 : 1 (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{3}$ (D) $\sqrt{\frac{3}{2}}$

19. The initial velocity of a particle is given by u (at t = 0) and the acceleration by f, where f = at (here t is time and a is constant). Which of the following relation is valid?

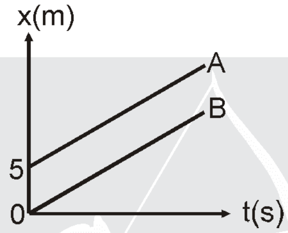
(A) $v = u + at^2$ (B) $v = u + \frac{at^2}{2}$ (C) $v = u + at$ (D) $v = u$

20. A student determined to test the law of gravity for himself walks off a sky scraper 320 m high with a stopwatch in hand and starts his free fall (zero initial velocity). 5 second later, superman arrives at the scene and dives off the roof to save the student. What must be superman's initial velocity in order that he catches the student just before reaching the ground ?

[Assume that the superman's acceleration is that of any freely falling body.] ($g = 10\ m/s^2$)

(A) 98 m/s (B) $\frac{275}{3}$ m/s (C) $\frac{187}{2}$ m/s (D) It is not possible

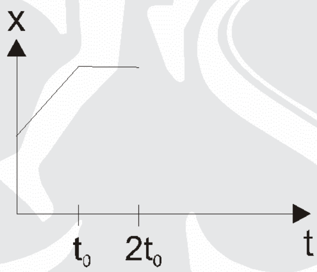
21. Figure shows position-time graph of two cars A and B. Lines are parallel.



(A) Car A is faster than car B. (B) Car A always leads Car B.

(C) Both cars are moving with same velocity. (D) Both cars have positive acceleration.

22. Figure shows the position time graph of a particle moving on the X-axis.



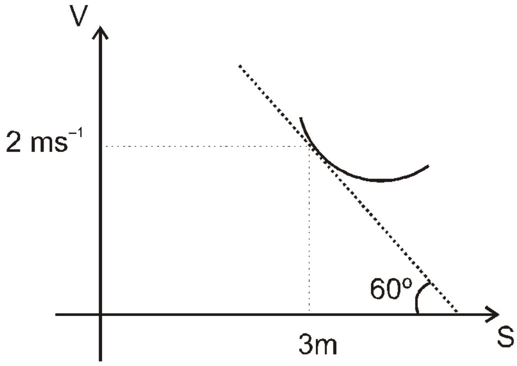
(A) the particle is continuously going in positive x direction

(B) area under x–t curve shows the displacement of particle

(C) the velocity increases up to a time t_o, and then becomes constant.

(D) the particle moves at a constant velocity up to a time t_o, and then stops.

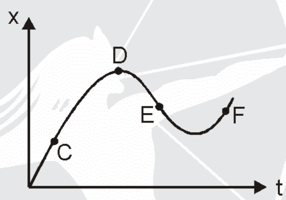
23. A particle is moving along a straight line whose velocity displacement curve is as shown in figure



What will be the acceleration of particle when it's displacement is 3m?

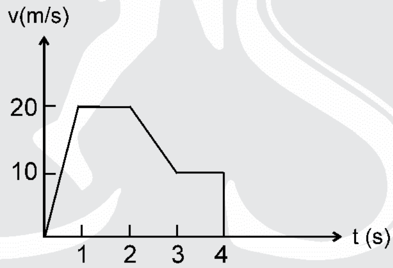
(A) $2\sqrt{3}$ ms^{-2} (B) $4\sqrt{3}$ ms^{-2} (C) $-2\sqrt{3}$ ms^{-2} (D) $\sqrt{3}$ ms^{-2}

24. In the displacement–time graph of a moving particle is shown, the instantaneous velocity of the particle is negative at the point :



(A) C (B) D (C) E (D) F

25. The variation of velocity of a particle moving along a straight line is shown in the figure. The distance travelled by the particle in 4 s is :



(A) 25 m (B) 30 m (C) 55 m (D) 60 m

ANSWER KEY

1.	(B)	**2.**	(D)	**3.**	(D)	**4.**	(B)	**5.**	(AC)
6.	(B)	**7.**	(A)	**8.**	(B)	**9.**	(C)	**10.**	(A)
11.	(D)	**12.**	(C)	**13.**	(B)	**14.**	(D)	**15.**	(B)
16.	(A)	**17.**	(D)	**18.**	(B)	**19.**	(B)	**20.**	(B)
21.	(BC)	**22.**	(D)	**23.**	(C)	**24.**	(C)	**25.**	(C)